

# Vamshidhar Reddy Mamidi

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## Objective

Aspiring Full-Stack Software Engineer passionate about building scalable, user-focused applications. Skilled in TypeScript, React, and backend development, with a strong foundation in APIs, cloud tools, and UI/UX design. Eager to contribute to cutting-edge AI products in high-impact environments.

## Technical Skills

- Languages: Java, Python, C++, JavaScript
- Frameworks/Technologies: AWS (Lambda, EC2, S3), React, Node.js, REST APIs
- Databases: MySQL, MongoDB, DynamoDB
- Tools: Git, Docker, Jenkins, Kubernetes
- Concepts: Object-Oriented Design, Distributed Systems, Data Structures & Algorithms, Agile Development
- AI & ML: Generative AI, Natural Language Processing (NLP), BERT, Content-Based Filtering, TF-IDF, Scikit-learn, PyTorch

## Education

Master of Science in Computer Science

Pace University, New York in May 2025

Relevant Coursework: Advanced Algorithms, Distributed Computing, Cloud Infrastructure, Machine Learning, Software Architecture

## Key Projects

Walmart USA Advanced Software Engineering Virtual Experience Program on Forage - July 2025

- Completed the Advanced Software Engineering Job Simulation where I solved difficult technical projects for a variety of teams at Walmart.
- Developed a novel version of a heap data structure in Java for Walmart's shipping department, showcasing strong problem-solving and algorithmic skills.
- Designed a UML class diagram for a data processor, considering different operating modes and database connections.
- Created an entity relationship diagram to design a new database accounting for all requirements provided by Walmart's pet department.

AWS Security Automation

- Developed AWS Config Rules using Python and CloudFormation to enforce secure S3 bucket policies.
- Created IAM roles and Lambda functions for real-time alerting on policy violations.
- Used Git and GitHub Actions to manage infrastructure and deploy automation workflows.

#### AI Learning Path Generator

- Developed an AI-powered personalized learning path recommendation system using python and machine learning algorithms.
- Developed a full-stack application using FastAPI (Python) and React that analyzes resumes using Generative AI.
- Implemented rule-based logic, content-based filtering (TF-IDF), and BERT-based models to optimize educational guidance based on user goals.
- Deployed using cloud infrastructure with Docker and GitHub Actions for CI/CD.
- Focused on clean UX design and rapid iteration cycles based on user feedback.

#### Certifications & Achievements

- AWS Certified Cloud Practitioner (2025)
- Walmart USA Advanced Software Engineering Virtual Experience Program on Forage - July 2025
- Hack Veda certification in Machine Learning with Python.